The Fixed-Block Cancellation Step in CPSV16 Theorem II.1

2026-05-11

1 Source passage

This note concerns the proof of CPSV16 Theorem `thm1`, Appendix MPV proof line 1182 of [CPGSV16]. The paper fixes a block $k_0 \in \{1, \dots, r_B\}$ and writes

$$\forall j, \langle V^{(N)}(B_{k_0}), V^{(N)}(A_j)\rangle \xrightarrow[N\to\infty]{} 0 \tag{1}$$

as a contradiction to eventual proportionality of the two full MPV families. Under the BNT decomposition, the proportionality hypothesis is projective proportionality: the scalar c_N is nonzero at the lengths used in the comparison, as recorded in `docs/paper-gaps/cpsv16_nonzero_proportionality_reading.tex`. Thus

$$\sum_j \mu_j^N \, V^{(N)}(A_j) = c_N \sum_k \nu_k^N \, V^{(N)}(B_k), \qquad c_N \neq 0, \tag{2}$$

and isolating the selected block gives

$$\sum_j \mu_j^N \, V^{(N)}(A_j) - c_N \nu_{k_0}^N \, V^{(N)}(B_{k_0}) = c_N \sum_{k \neq k_0} \nu_k^N \, V^{(N)}(B_k). \tag{3}$$

Corollary `Lem1` in [CPGSV16] says that, from the left-hand decay, the family

$$\dim \operatorname{span}\left(\left\{V^{(N)}(A_j)\right\}_{j=1}^{r_A} \cup \left\{V^{(N)}(B_{k_0})\right\}\right) = r_A + 1 \tag{4}$$

for all sufficiently large N. Equivalently, the displayed family is linearly independent, so the selected block cannot be absorbed by the others without contradiction. [CPGSV16, Appendix MPV proof, line 1182]

2 Formal boundary

The formal boundary in [CPGSV16] is narrower:

$$\forall^{\mathrm{cof}} N, \dim \operatorname{span}\left(\left\{V^{(N)}(A_j)\right\}_{j=1}^{r_A} \cup \left\{V^{(N)}(B_{k_0})\right\}\right) = r_A + 1. \tag{5}$$

The surviving SectorBNT API does not currently contain this single-block linear-independence formulation. It supplies the stronger full-family mechanism MPSTensor.IsBNTCanonicalForm.combined_family_eventually_li, under all-pairwise cross-overlap decay; this is sufficient for the coefficient-comparison contradiction, but it is not the literal single-block statement displayed above. The exact fixed-block contradiction step is still open:

$$(\forall j, \langle V^{(N)}(B_{k_0}), V^{(N)}(A_j)\rangle \to 0) \Longrightarrow \neg \left(\sum_j \mu_j^N \, V^{(N)}(A_j) = c_N \sum_k \nu_k^N \, V^{(N)}(B_k) \right). \tag{6}$$

No current Lean declaration carries the retired fixed-right or fixed-left one-copy fixed-block names. The mathematical target remains the CPSV fixed-block cancellation at [CPGSV16, Appendix MPV proof, line 1182], now to be stated on the SectorBNT sector surface.

The earlier conditional helpers requiring combined-family linear independence (`_phase_sum_li`) and the residual-span variants have been retired in the 2026-05-13 audit; see `audits/2026-05-13_cpsv16_ft_paper_vs_code_structural_map.md`. Those helpers should not be used as replacements for the fixed-block sentence at Appendix MPV proof line 1182, because the needed combined-family independence is not supplied by Corollary `Lem1`. The remaining open obligation is the fixed-block cancellation itself, to be discharged without combined-family LI or residual-span exclusion.

Retirement note: this note intentionally no longer lists the retired no-tail, leading-only, partner-identification, leading-tail, residual-span, or `_phase_sum_li` declarations. They were deleted as wrong-direction or orphan scaffolding in the same audit.

3 Current discharge plan

The current plan is recorded in `audits/2026-05-13_cpsv16_ft_sorry_discharge_plan.md`. Plan A is to prove an exact eventual leading-coefficient identity

$$c_N \, \mu_0^{B\,N} \, \zeta^N = (\mu_0^A)^N \tag{7}$$

for all sufficiently large N from the proportionality identity and the BNT linear independence on the CPSV fixed-block family. Once this identity is available, the proportional case can clone the equal-MPV tail subtraction and induction architecture, without assuming linear independence of all residual blocks on both sides.

3.1 Factored analytic discharge and the lower-bound gap

Classification: Scope restriction.

The intended factored analytic discharge uses four explicit analytic hypotheses:

1. unit-modulus sector weights on both the A-family and the B-family;
2. off-diagonal cross-overlap decay on the fixed-block family, i.e., $\langle V^{(N)}(B_{k_0}), V^{(N)}(B_k)\rangle \to 0$ for $k \neq k_0$;
3. a nonzero limit of the self-overlap at the fixed block, $\langle V^{(N)}(B_{k_0}), V^{(N)}(B_{k_0})\rangle \to \ell \neq 0$; and
4. an eventual positive lower bound $\|c_N\| \geq \delta > 0$.

Hypotheses (1)–(3) are present in the source via the BNT structure [CPGSV16, Theorem `thm1`]. Hypothesis (4) is not directly supplied: the cited argument rules out cancellation by appeal to the dominant-adjusted scalar limit, but deriving the explicit lower bound $\|c_N\| \geq \delta$ from the structural data of the canonical form is not yet formalized and is recorded as an open obligation in `audits/2026-05-13_cpsv16_ft_sorry_discharge_plan.md`. The earlier non-cancellation helpers carrying this lower-bound hypothesis were part of the retired Tier 3 route and should not be cited as present declarations.

References

[CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: Annals of Physics 378 (2017), 100–149, 2016.